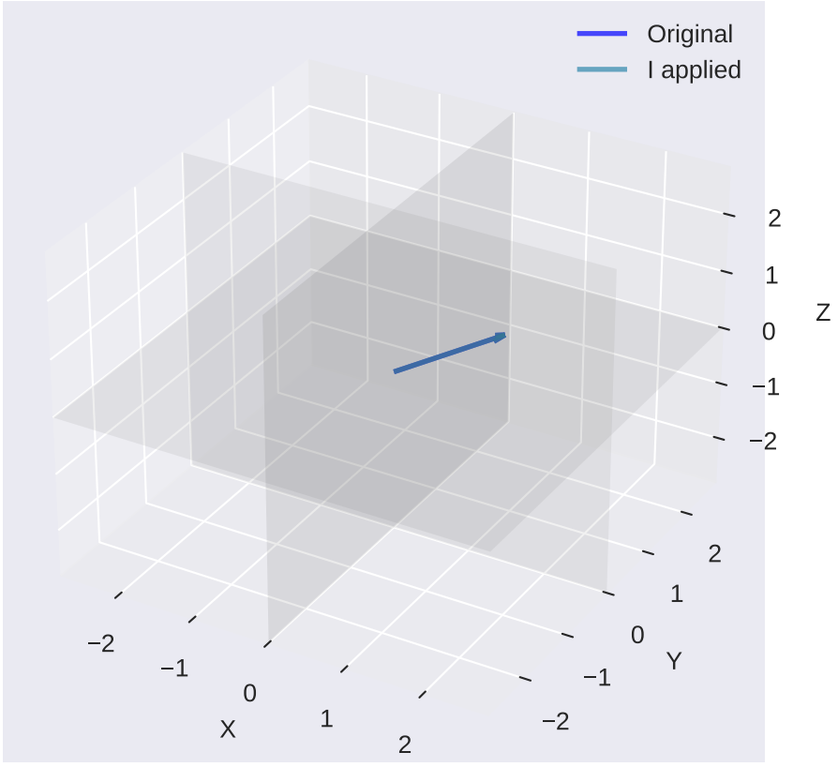
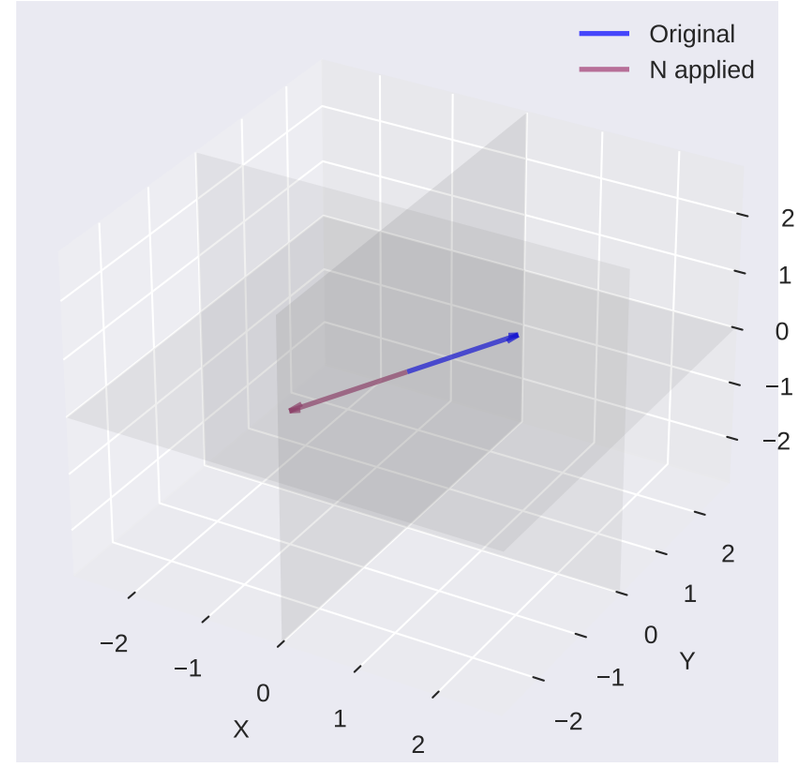
3D Vector Transformations by Klein-4 Operators
Original vector (first 3 dimensions): [0.43368507 1.8750499 -0.41055209]

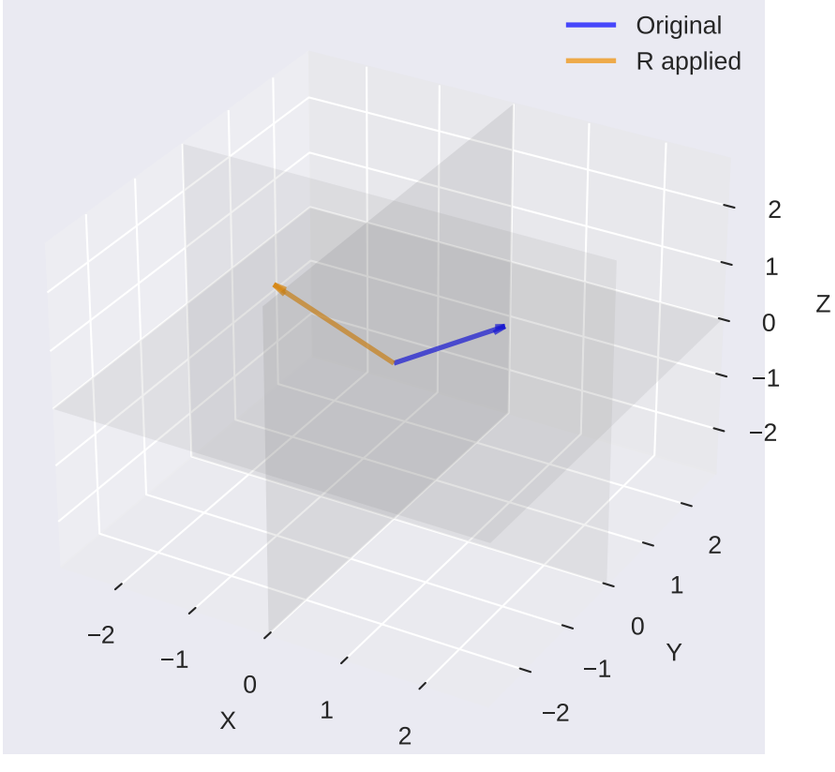
Operator I



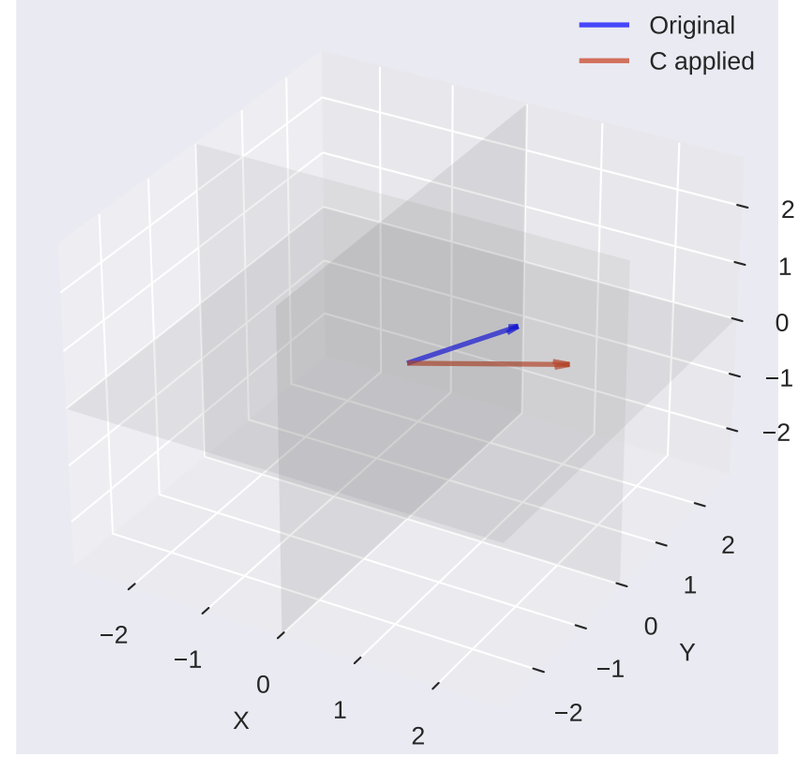
Operator N



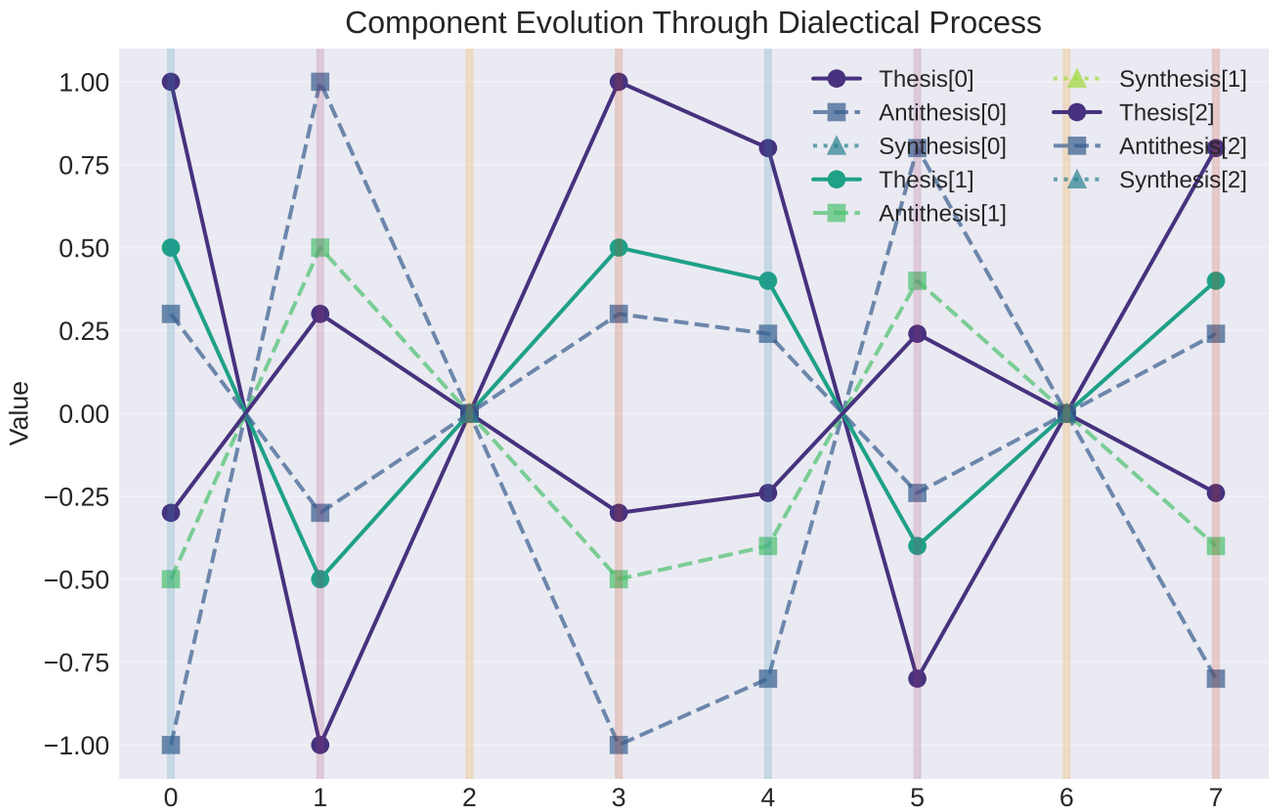
Operator R

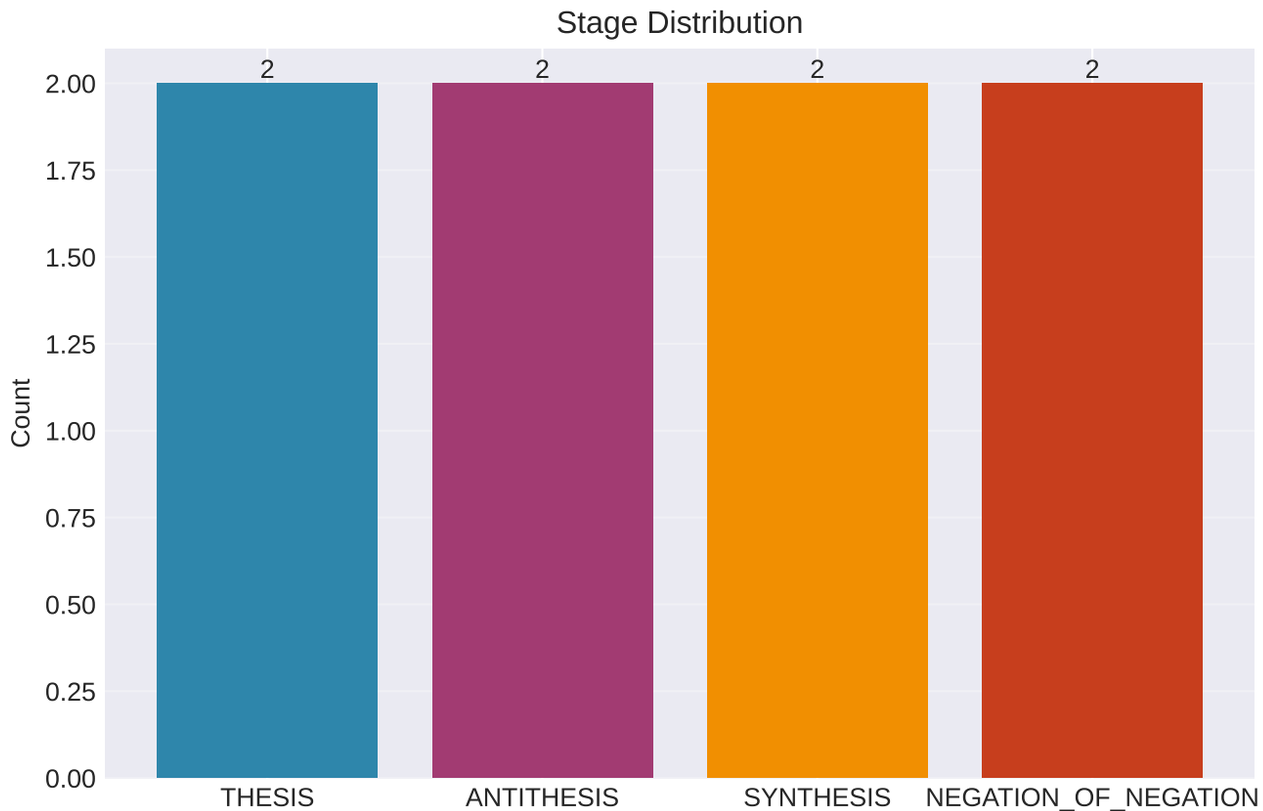


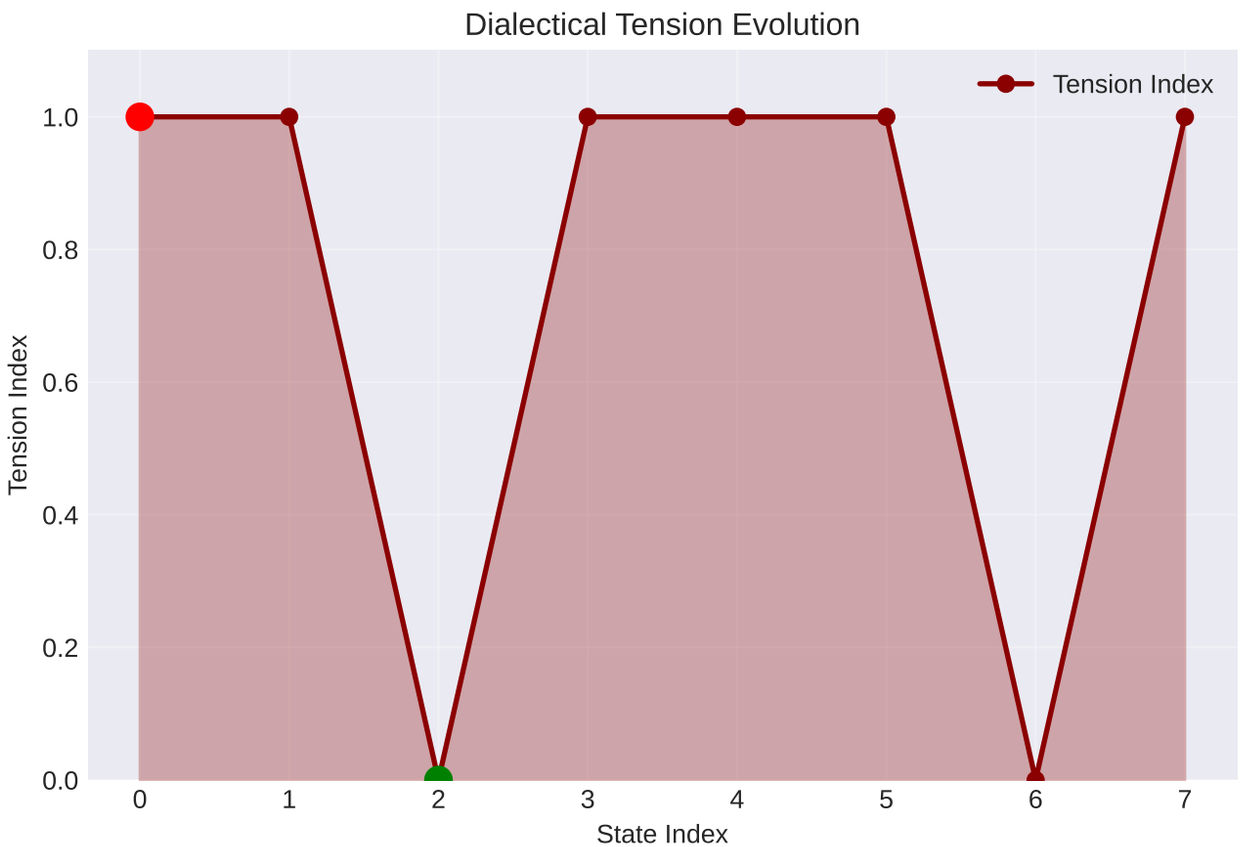
Operator C

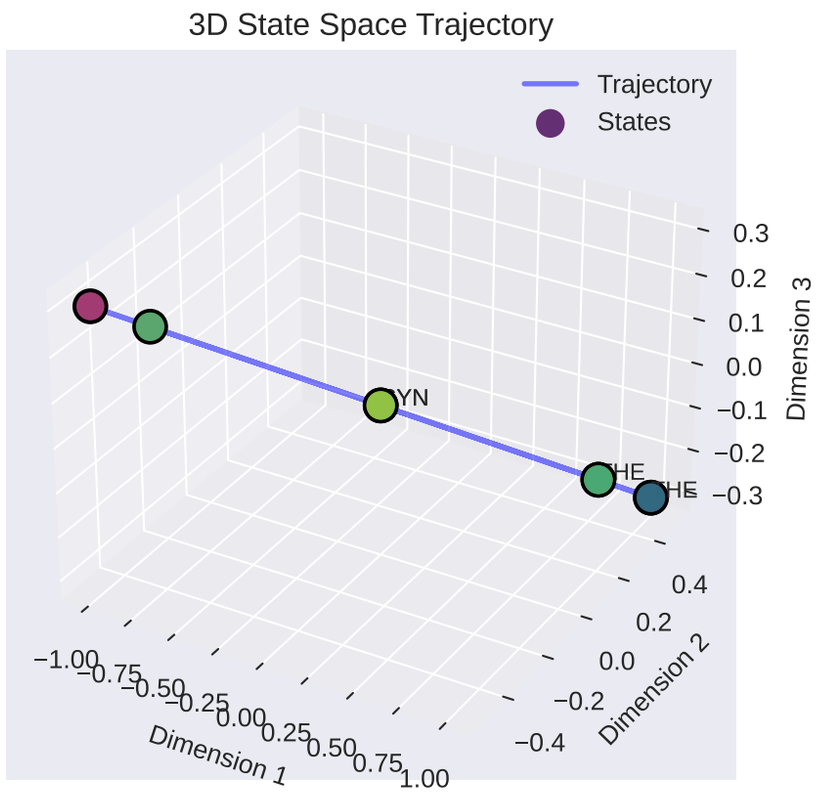


Dialectical Process Evolution
Initial thesis: [1.0, 0.5, -0.3]... | Cycles: 2 | Synthesis method: DIALECTICAL

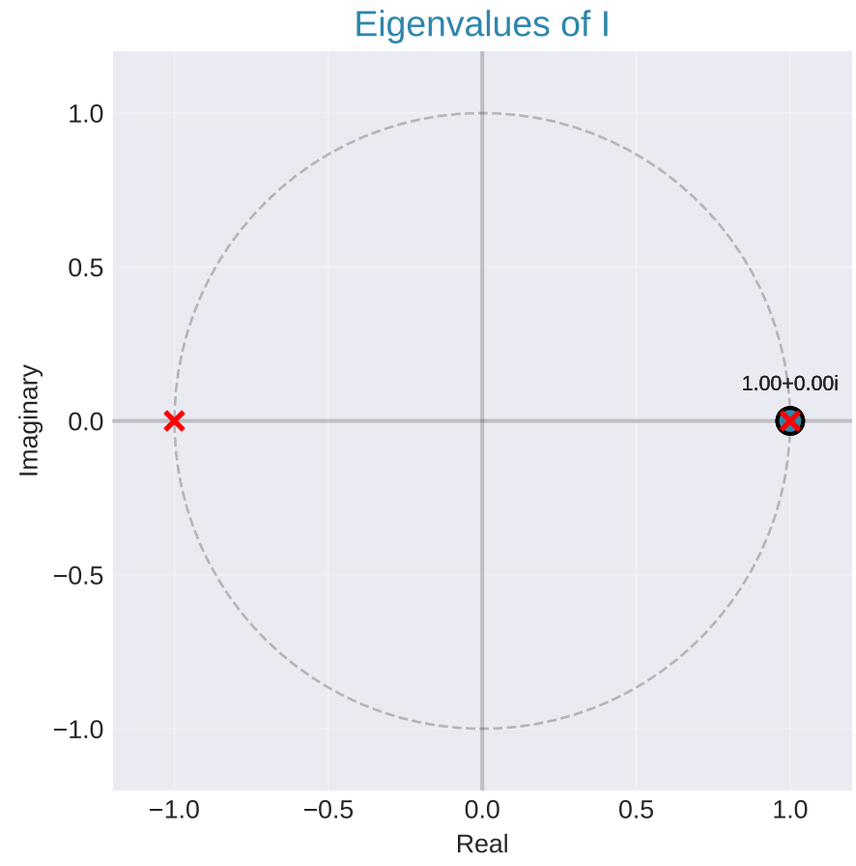


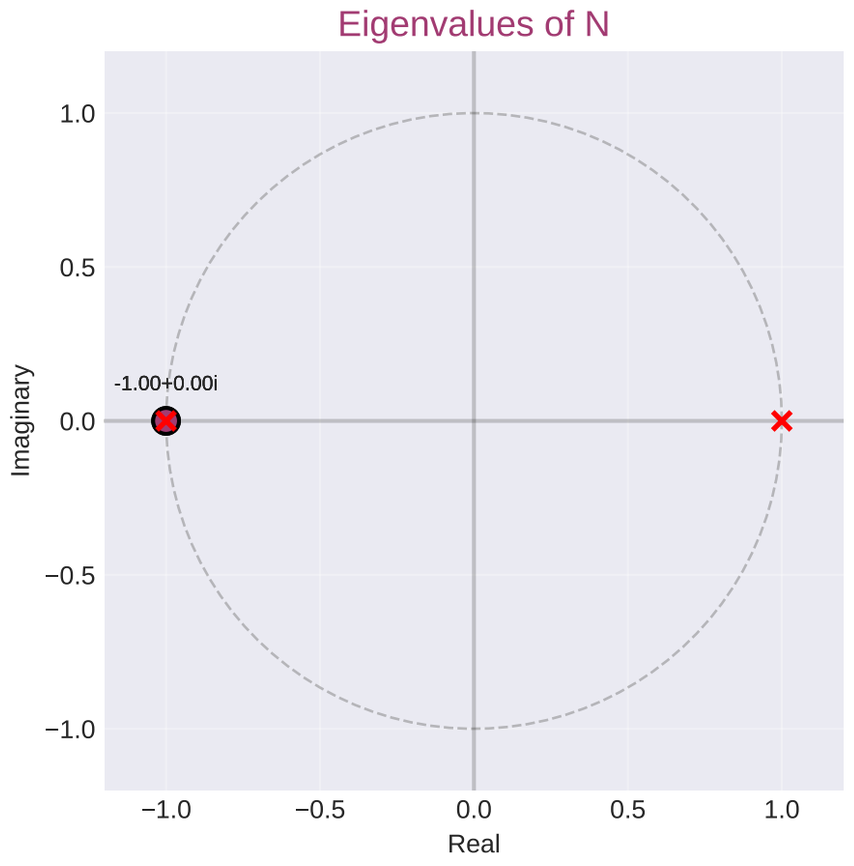


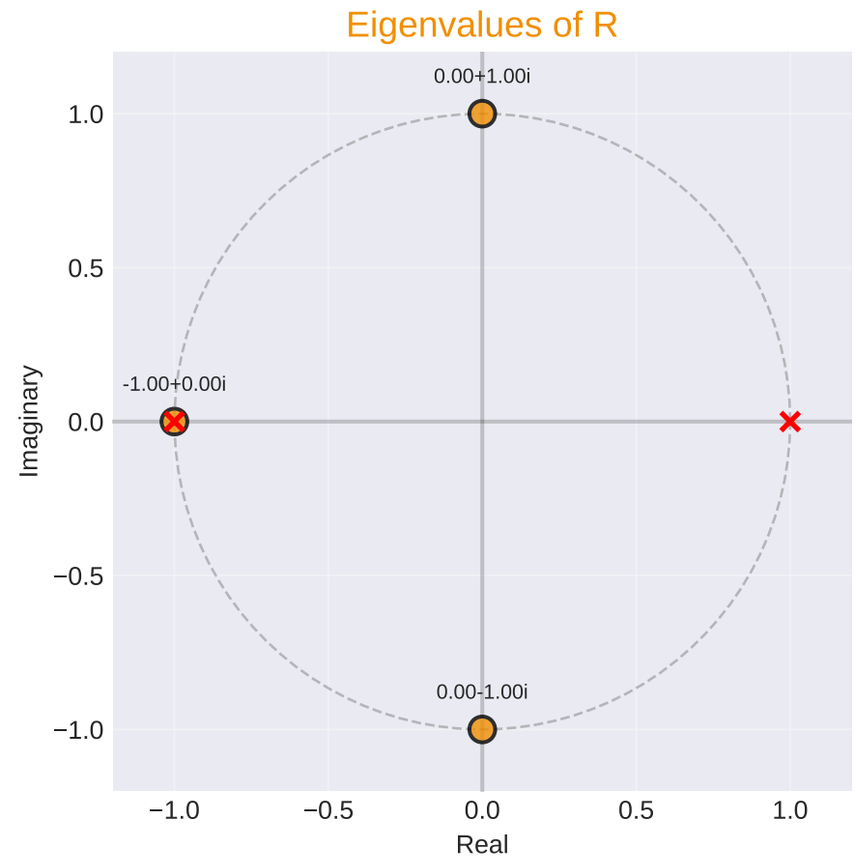


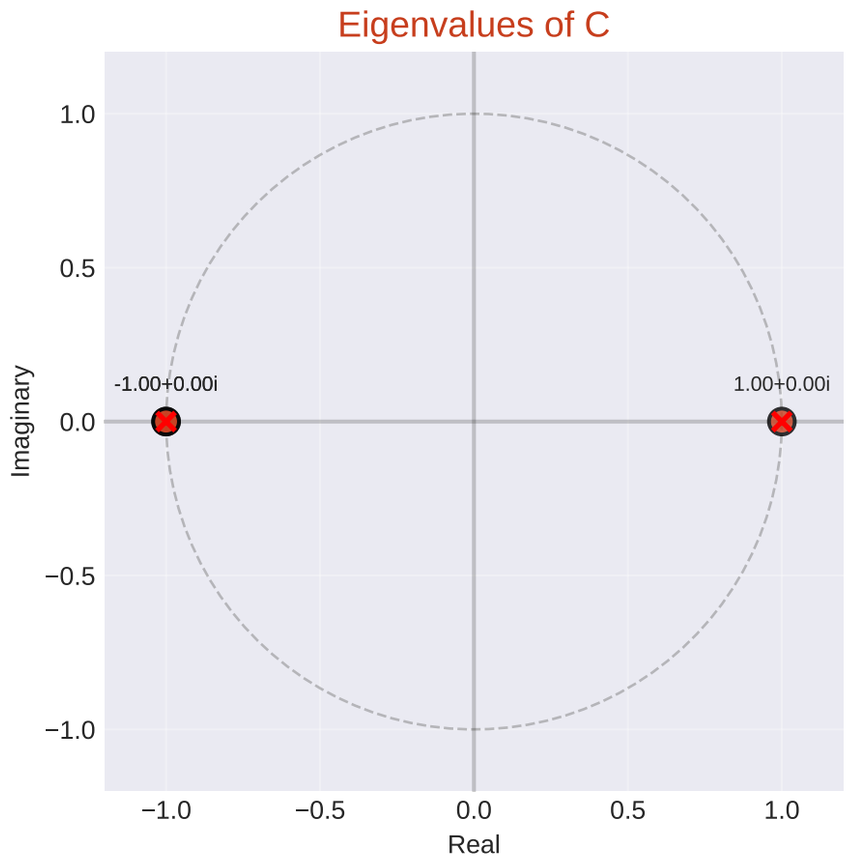


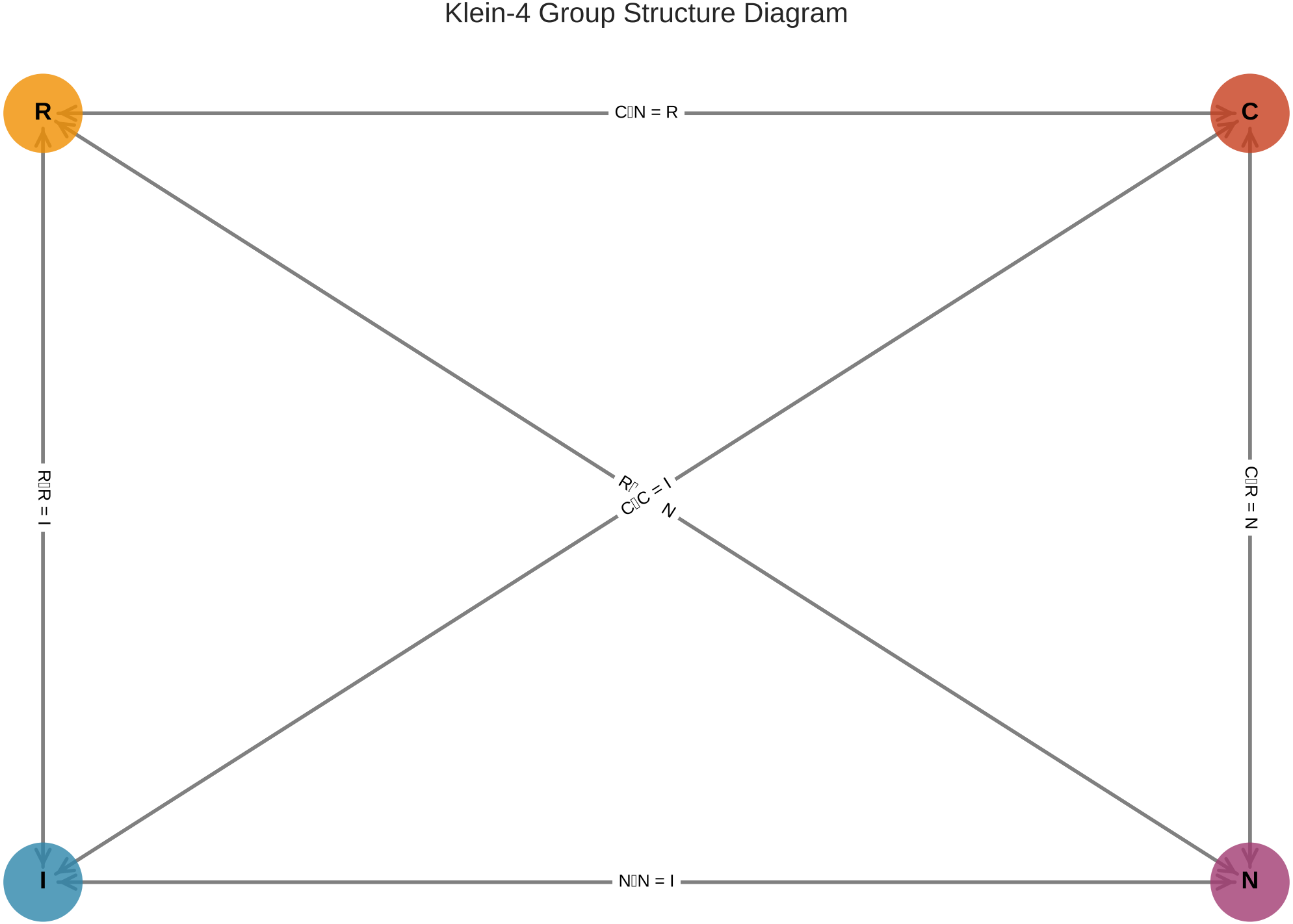
Eigenvalues on Complex Plane (Should be ±1)



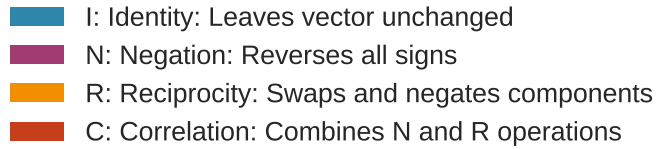






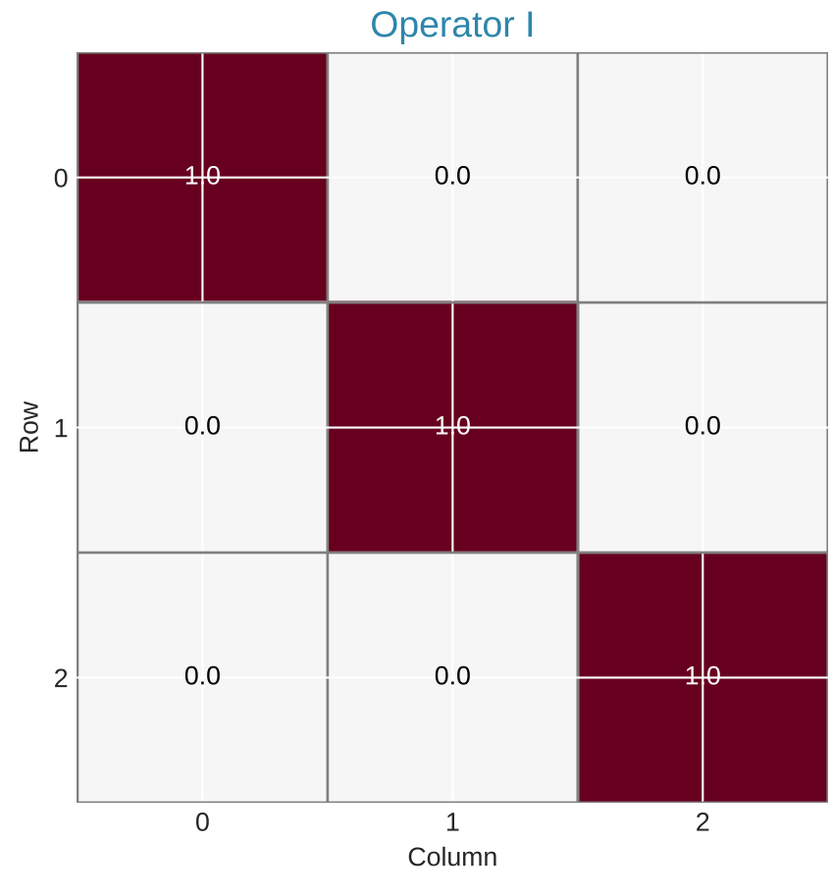

Klein-4 Group Structure Diagram
R
C
I
N
C·N = R
R·R = I
R·C = I
C·C = I
C·R = N
N·N = I

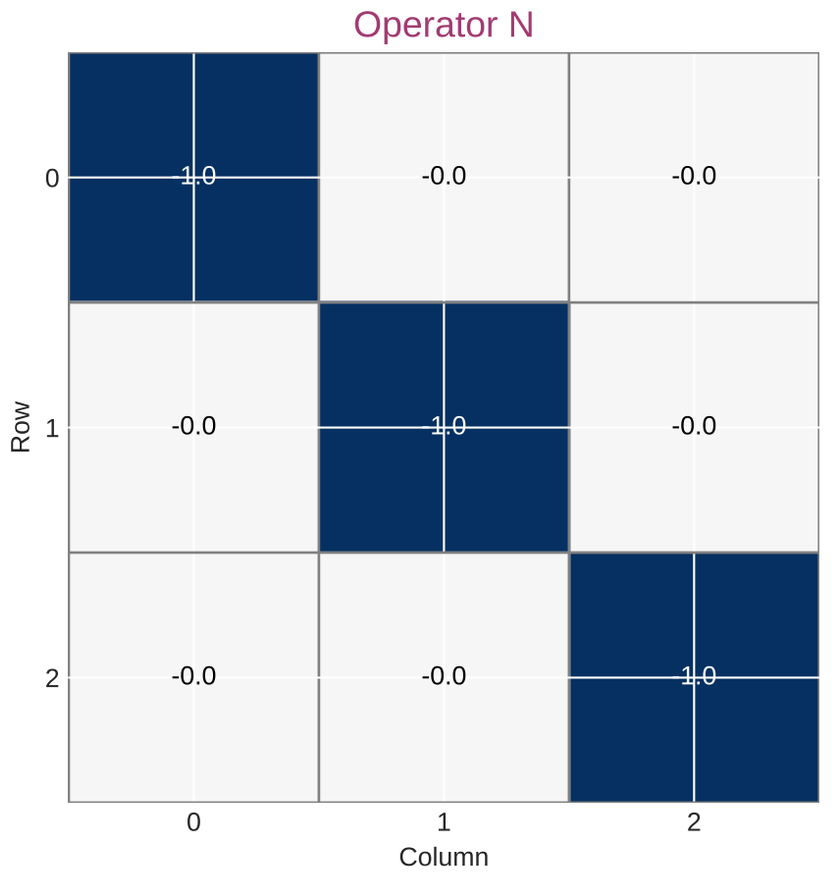
Klein-4 Group Cayley Table

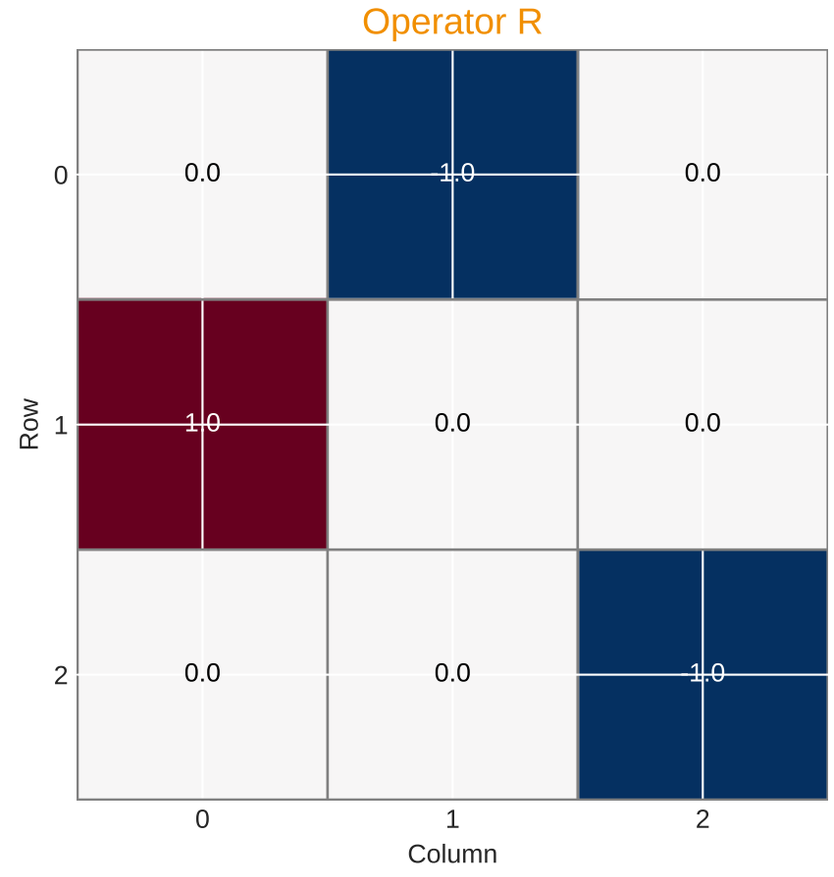


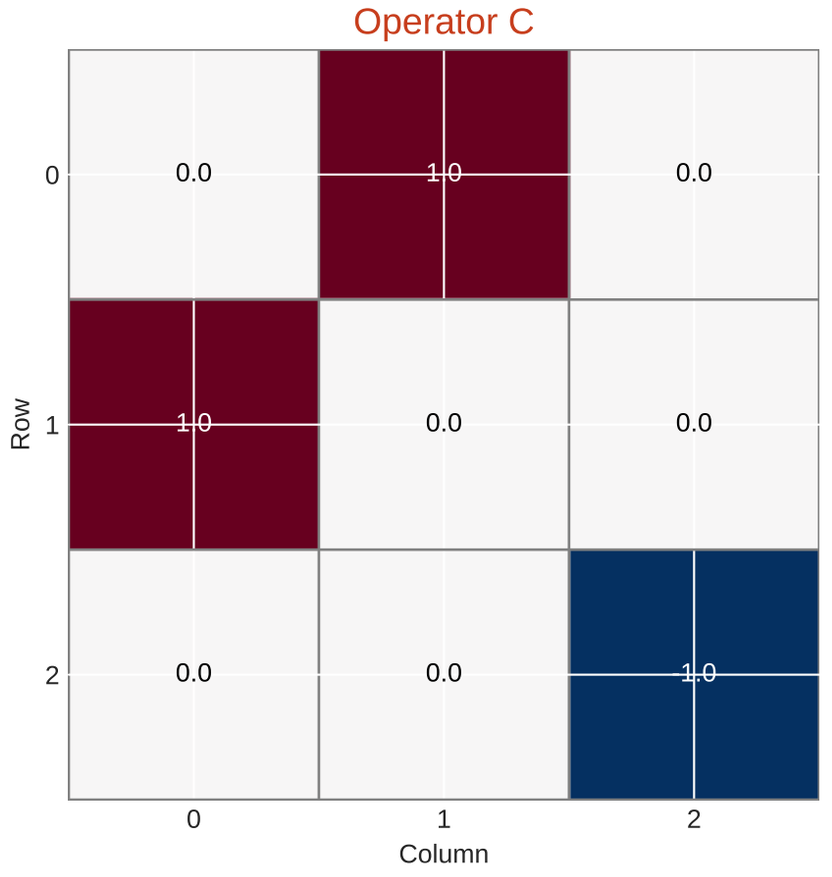
	I	**N**	**R**	**C**
I	I	N	R	C
N	N	I	C	R
R	R	C	I	N
C	C	R	N	I

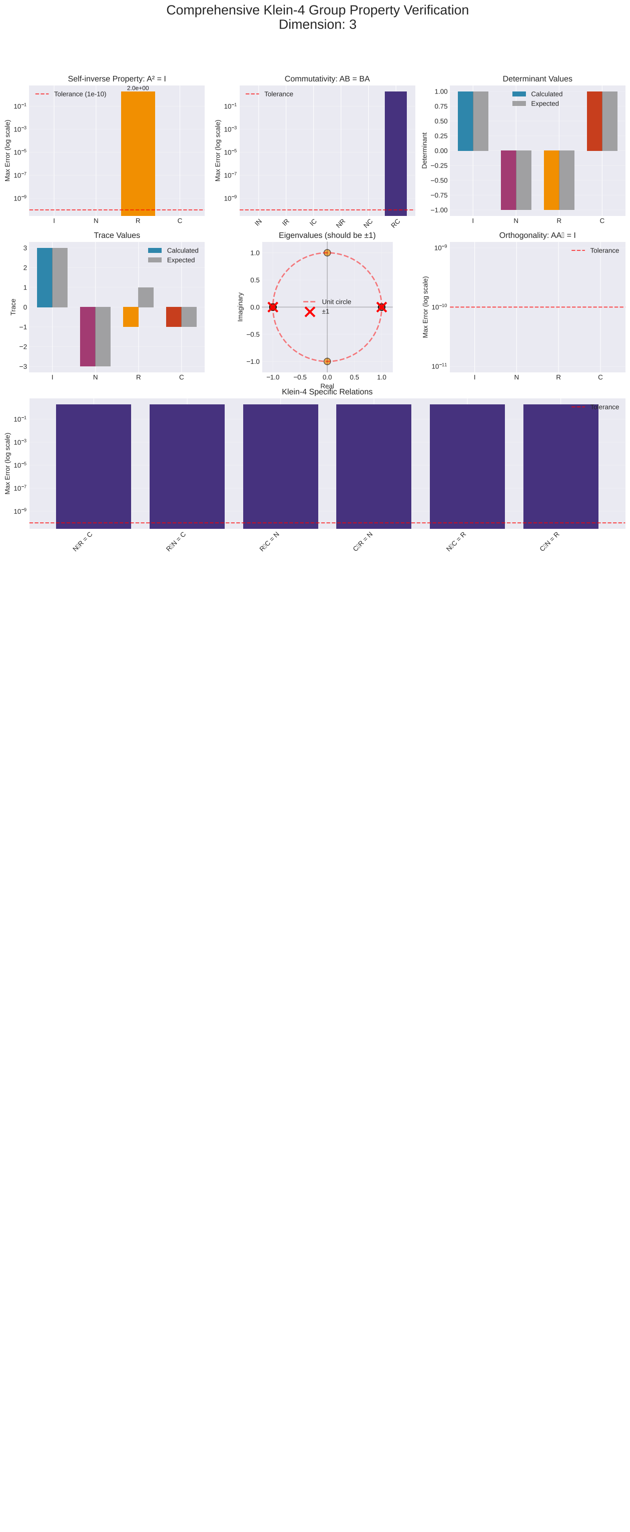
Klein-4 Operator Matrices (Dimension 3)





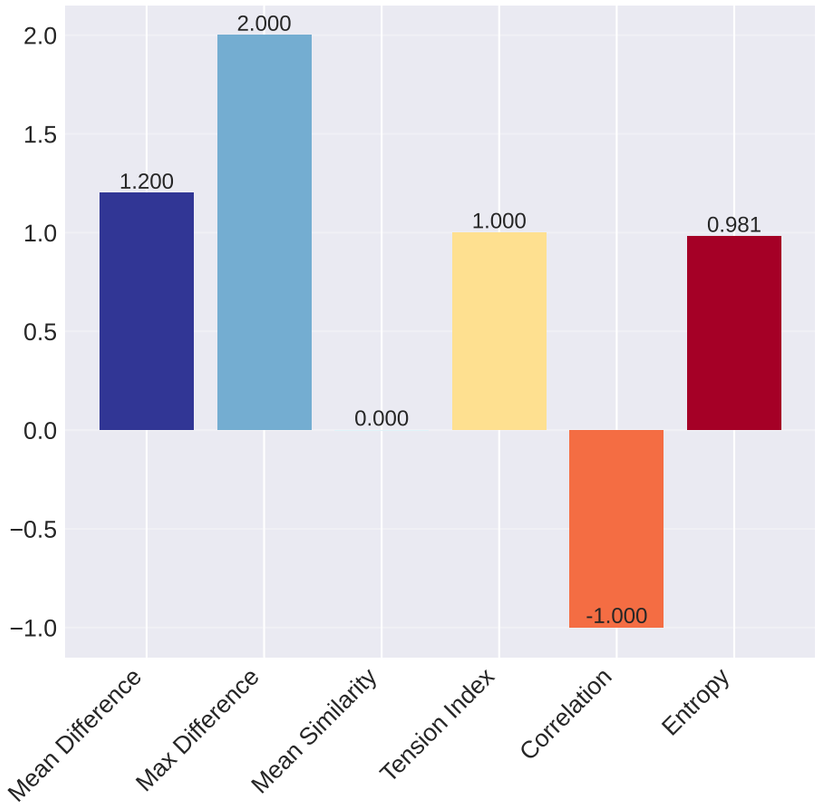




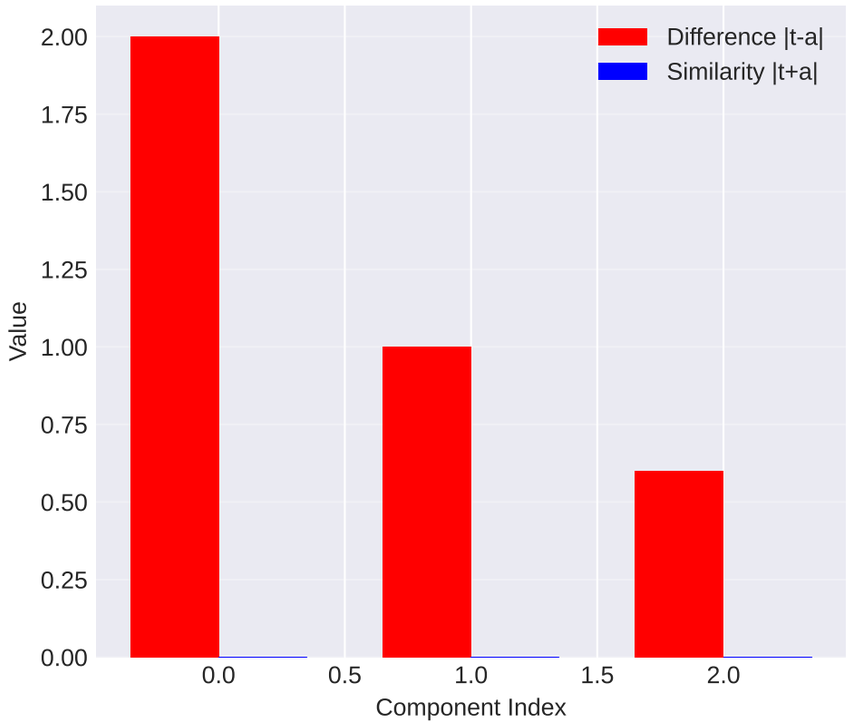

Comprehensive Klein-4 Group Property Verification
Dimension: 3
Self-inverse Property: A² = I
Tolerance (1e-10)
2.0e+00
Max Error (log scale)
Commutativity: AB = BA
Tolerance
IN
IR
IC
NR
NC
RC
Determinant Values
Calculated
Expected
Determinant
Trace Values
Trace
Eigenvalues (should be ±1)
Unit circle
±1
Real
Imaginary
Orthogonality: AA = I
Klein-4 Specific Relations
0.0e+00
0.0e+00
0.0e+00

Detailed Tension Analysis
Thesis: [1.0, 0.5, -0.3]... | Antithesis: [-1.0, -0.5, 0.3]...

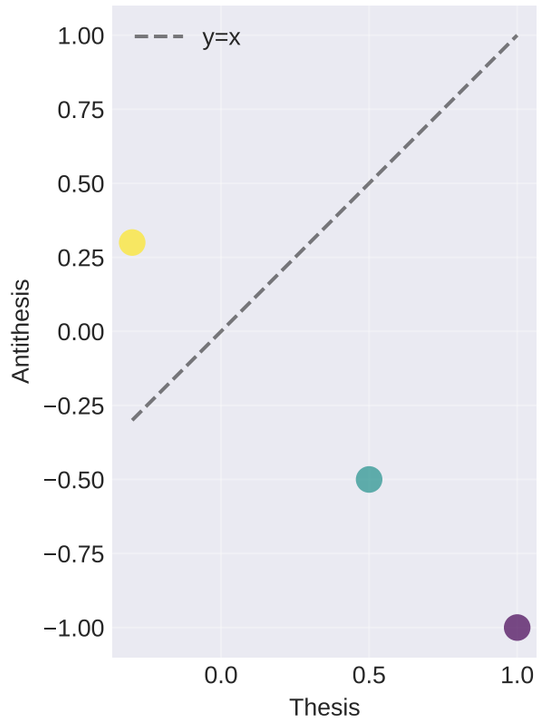
Tension Components



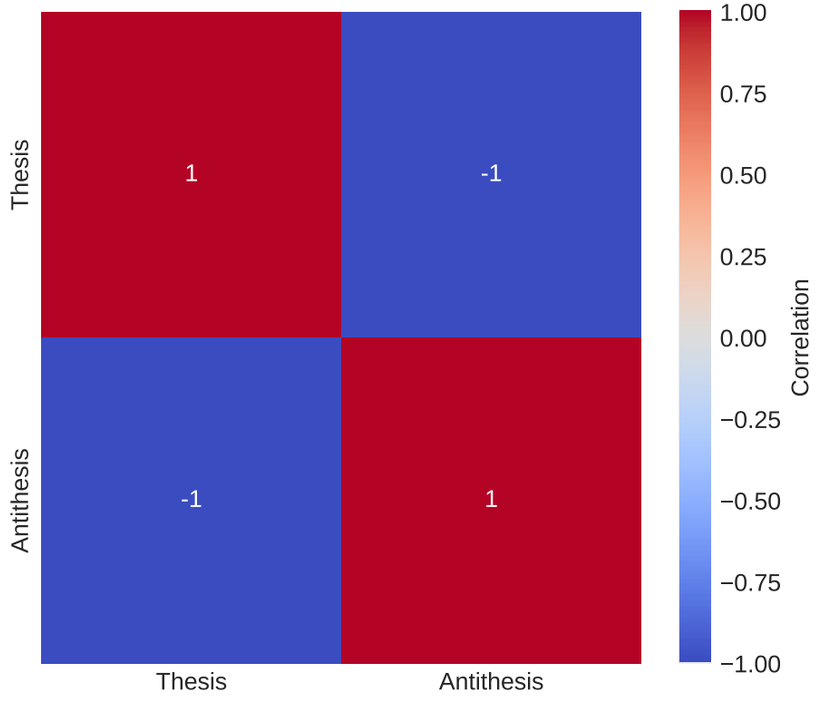
Component-wise Comparison



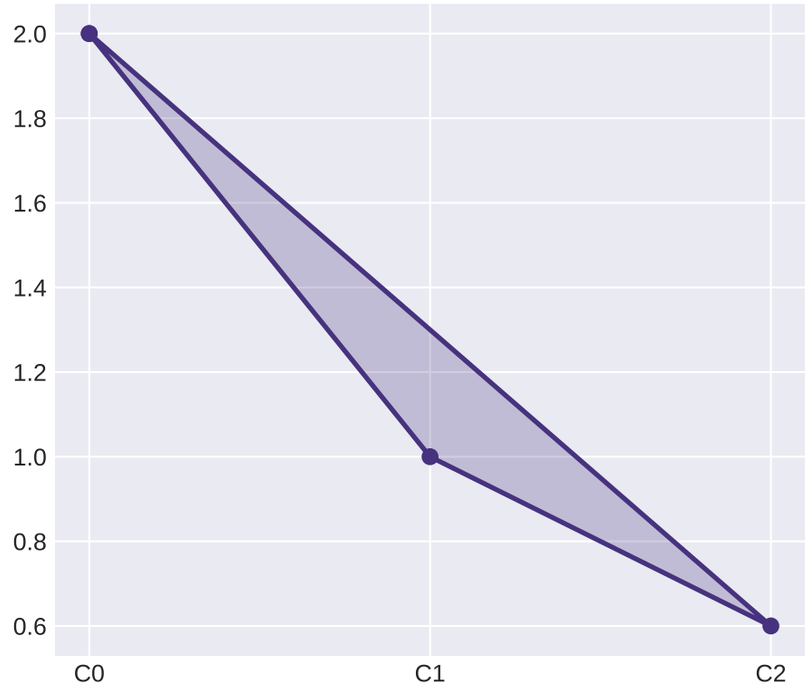
Thesis vs Antithesis



Correlation Matrix: -1.000



Radial Difference Plot



TENSION ANALYSIS SUMMARY

Stage: THESIS
Dimensions: 3
Tension Index: 1.000
Mean Difference: 1.200
Max Difference: 2.000
Correlation: -1.000
Entropy: 0.981
Has Synthesis: False

HIGH TENSION: Strong opposition